

COURSE TITLE : ANALYTICAL AND DIAGNOSTIC EQUIPMENTS
COURSE CODE : 4241
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Electronic instruments and measurements	14
2	Clinical Lab Instruments	14
3	Blood Pressure and Blood Flow measurements	14
4	Biomedical recorders	14
TOTAL		56

Course General Outcome :

MODULE	G.O	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To understand the terminologies in measurements
	2	To understand the working of display and recording devices
2	3	To understand the working of different Clinical Lab Instruments
	4	To understand the principle of electrochemical method of pH measurement
	5	To understand the principle of pO ₂ and pCO ₂ measurement
3	6	To Know the different methods of Blood Pressure and temperature measurements
	7	To know the working of blood flow meters and Coulter counter
4	8	To understand the different Biomedical recorders
	9	To understand ECG and EMG recorders
	10	To comprehend the recording procedure of EEG and its applications

G.O.: General Outcome

On completion of the study the student will be able:

MODULE I ELECTRONIC INSTRUMENTS AND MEASUREMENTS

1.1.0 To understand the terminologies in measurements

- 1.1.1 To define instrument accuracy, precision, sensitivity, resolution and error
- 1.1.2 To differentiate between analogue and digital meters
- 1.1.3 To explain the working of a galvanometer
- 1.1.3 To explain with block diagram of digital multi meter

1.2.0 To understand the working of display and recording devices

- 1.2.1 To explain the principle and working of analog and digital oscilloscopes.
- 1.2.2 To describe the measurement of voltage (DC & AC), frequency, phase angle and time period with CRO

- 1.2.3 To explain the working of X-Y Recorders,
- 1.2.4 To explain the working of circular Chart Recorders
- 1.2.5 To explain the working of Strip Chart Recorders

MODULE II CLINICAL LAB INSTRUMENTS.

2.1.0 To understand the working of different Clinical Lab Instruments.

- 2.1.1 To state and derive Beer Lamberts law.
- 2.1.2 To differentiate the monochromators used in clinical lab instruments.
- 2.1.3 To explain the principle and working of colorimeters and spectrophotometers
- 2.1.4 To explain the principle and working of flame photometers
- 2.1.5 To explain Haemoglobin meter and Oximeters
- 2.1.6 To explain the block diagram of auto analyzer.

2.2.0 To understand the principle of electrochemical method of pH measurement

- 2.2.1 To explain the principle of pH measurement.
- 2.2.2 To draw electrodes for pH measurement.
- 2.2.3 To describe the block diagram of pH meter

2.3.0 To understand the principle of pO₂ and pCO₂ measurement

- 2.3.1 To explain the principle and construction of electrode for measurement of PO₂
- 2.3.2 To explain the principle and construction of electrode for the measurement of PCO₂
- 2.3.3 To explain catheter tip electrodes for the measurement PO₂ & PCO₂
- 2.3.4 To describe the block diagram of blood gas analyser.

MODULE III BLOOD PRESSURE AND BLOOD FLOW MEASUREMENTS

3.1.0 To know the different methods of Blood Pressure and temperature measurements.

- 3.1.1 To classify blood pressure into arterial and ventricular BP.
- 3.1.2 To explain non invasive methods of BP measurement with principle of sphygmomanometer.
- 3.1.3 To describe various transducers used for invasive BP measurement.
- 3.1.4 To define pulse and respiratory sensors.
- 3.1.5 To describe various transducers used for temperature measurement.

3.2.0 To know the working of blood flow meters and Coulter counter

- 3.2.1 To describe the principle and working of electromagnetic blood flow meters
- 3.2.2 To describe the principle and working of ultrasonic blood flow meters.
- 3.2.3 To describe the working of Coulter blood cell counter.

MODULE IV BIOMEDICAL RECORDERS

4.1.0 To understand the different biomedical recorders

- 4.1.1 To explain the basic blocks present in a bio-signal sensing and recording systems.
- 4.1.2 To explain different types of electrical isolation techniques used in biomedical devices.

4.1.3 To explain characteristics and requirements of amplifiers used in biomedical devices.

4.2.0 To understand ECG and EMG recorders

4.2.1 To illustrate the lead configuration for recording of ECG

4.2.2 To describe the working of basic ECG recorder

4.2.3 To explain microprocessor based three channel ECG recorder

4.2.4 To explain Stress ECG, different exercise protocols used.

4.2.5 To explain Vector cardiography and Holter Cardiography

4.2.6 To describe the working of electromyograph and its applications.

4.3.0 To comprehend the recording procedure of EEG and its applications

4.3.1 To explain electrode placement for EEG recording.

4.3.2 To explain the working of EEG recorder and applications of EEG.

CONTENTS

MODULE I ELECTRONIC INSTRUMENTS AND MEASUREMENTS

Definitions of instrument accuracy, precision, sensitivity, resolution and error -

Analogue and digital meters - Working of a galvanometer - Digital multi meter

Display and recording devices

Working of analog and digital oscilloscopes - Measurement of voltage (DC & AC), frequency, phase angle and time period with CRO - X-Y Recorders - circular Chart Recorders - Strip Chart Recorders

MODULE II CLINICAL LAB INSTRUMENTS

Derivation of Beer Lamberts law - Monochromators – Working principle of colorimeters , spectrophotometers ,flame photometers - Haemoglobin meter and Ox meters - block diagram of auto analyzer.

Electrochemical method of pH measurement

Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter

Principle of pO₂ and pCO₂ measurement

Principle and construction of PO₂ electrode and PCO₂ electrode - catheter tip electrodes for measuring PO₂ & PCO₂ - block diagram of blood gas analyser

MODULE III BLOOD PRESSURE AND TEMPERATURE MEASUREMENTS

Arterial and ventricular BP - non invasive methods of BP measurement - principle of sphygmomanometer - Pressure Transducers in invasive BP measurement – Definition of pulse and respiratory sensors - temperature transducers

Blood flow meters and Coulter counter

Working principle of electromagnetic blood flow meters and ultrasonic blood flow meters - Working of Coulter blood cell counter

MODULE IV BIOMEDICAL RECORDERS

Block diagram of bio-signal sensing and recording systems - electrical isolation techniques - characteristics and requirements of amplifiers in recorders.

ECG and EMG recorders

ECG lead system - working of ECG recorder – working of microprocessor based three channel ECG recorders - Stress ECG - Exercise protocols - working of electromyograph - applications

EEG Recorders

10-20 lead system EEG recording - working of EEG recorder - applications of EEG

TEXT BOOKS

- | | | |
|---|---|--|
| 1. Handbook of Analytical Instruments | - | R S Khandpur-Tata McGraw Hill-2006 |
| 2. Handbook of Biomedical Instruments | - | R S Khandpur- Tata McGraw Hill-2006 |
| 3. Introduction to Biomedical Instrumentation | - | Arumugham-Anuradha Publications |
| 4. Electronics Instrumentation | - | A K Sawhney, Puneet Sawhney,
Dhanpat Rai & Co. 2008 |

REFERENCES

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|---|--|--|
| 1. Biomedical Instrumentation and Measurements | - | Leslie Cromwell-Prentice Hall-2011 |
| 2. Principles of Applied Biomedical Instrumentation | - | Geddes and Baker-John Wiley-1989 |
| 3. Principles of Biomedical Instrumentation - | Richard Aston-Merrill Publishing Company, 1990 | |
| 4. Medical Physics | - | John R Cameron, J G Skofronick, John Wiley & Sons-
1997 |